# Response to the reviewer (fifth round)

A reliability-constrained, multi-objective framework for minimum-mass design of steel frame structures under material and loading uncertainty

[Author name]

We thank the reviewer for the final comments. No new computations were made, and all numerical results are unchanged. Section and table numbers refer to the revised manuscript.

**Comment 2 (labelling of the $t_2$ sensitivity result).** *Identify 3.16 explicitly as the $t_2$-model point estimate, unless a lower confidence bound has been calculated.*

**Response.** No confidence interval or lower bound was computed for the $t_2$ model; 3.16 is a point estimate. This is now stated explicitly wherever the value appears:

- Section 3.3: the $t_2$ model is introduced as a further case "for which only point estimates are reported", and the change of the indices under $t_2$ refers to point estimates.
- Section 4.6: "With the more extreme $t_2$ residual model, the point estimate is 3.16; no confidence interval or lower bound was computed for the $t_2$ model." The later comparison with the $t_3$ lower bound also says "the $t_2$ point estimate is 3.16".
- Conclusion 5: "Under the more extreme $t_2$ residual model, the point estimate is 3.16 (no lower bound was computed for that model)."
- Table 6: the column headings, generated by `make_figures.py`, now read "$\hat{\beta}$ [model-cond. 95% CI] ($\beta^{lcb}$), $t_3$" and "$\hat{\beta}$ (point est.), $t_2$".

The Abstract does not quote the $t_2$ value. It describes the complete-model result as resting on an assumed heavy-tailed residual model, with a model-conditional interval.

**Comment 3 (limited meaning of "exact superposition").** *Ensure that "exact" refers to the linear-response superposition step, not to the whole surrogate.*

**Response.**

- Introduction, contribution 1: "an exact superposition surrogate" is now "a superposition surrogate". The item adds that the superposition of the linear unit-case responses is exact within the linear analysis, whereas the $B_1$–$B_2$ amplification approximates the full second-order response, with a reference to Section 3.3.
- Section 2.2: a new paragraph after Eq. (1) states that the equation is exact for the first-order linear analysis only. It says that this exactness does not extend to the second-order response, because the $B_1$–$B_2$ amplification is an approximation of the full P-$\Delta$ analysis whose error is quantified in Section 3.3.
- Section 3.3 and the Discussion already describe the drift discrepancy and the unconservative three-storey result as errors of this approximation. No "exact" wording refers to the surrogate there.

**Comment 4 (final submission preparation).**

**Response.** The author and affiliation placeholders remain, because the names and affiliations have not yet been supplied to us. They will be inserted before submission. The manuscript keeps the exact repository commit that produced it. We have also retained the qualifications the reviewer lists: the paper does not claim that reliability-based optimisation saves steel, that the genetic algorithm proves global optimality, or that the model-conditional tail estimate is a distribution-free guarantee.